MZUZU UNIVERSITY

Faculty of Information Science and Communications

Department of Information and Communication Technology

Programme(s)	:	Bachelor of Science in Information and Communication Technology
Subject	:	Information and Communication Technology
Course Title	:	Introduction to Programming
Course Code	:	ICT 1102
Duration	:	14 Weeks
Lecture Hours/Week	:	Four
Tutorial/Semester Hours	:	Two
Practical	:	Two
Assessment	:	40% Continuous Assessment 60% Final Examination

Aims of the Course

The aim of the course is to provide students with a thorough understanding in the key concepts, techniques and methods in programming.

Course Objectives

By the end of this course students should be able to:

- Describe developments in programming languages and reasons for their evolution
- Apply appropriate ways to control and manipulate data.
- Break down data into structures appropriate for the problem to be addressed
- Formulate data into structures appropriate for the problem to be addressed
- Utilize a range of common programming concepts as building blocks in the creation of structured programs
- Define code requirements from the models
- Apply a range of techniques to modeling systems as objects
- Apply a range of testing approaches to check the correctness of programming code and explain other evolving ways of designing and building programs.
- Describe the importance of documentation in software development

Topics of Study

History of programming
- Describing Programming language
- Language Models
- Chronology of Programming
- Object Oriented Concepts
- Classes
- Encapsulation (or Information Hiding) and Abstraction
- Messages and Operations
- Relationships
- Polymorphism

Variables, Control Structures and Calculations
- Variable Types and Names
- Calculations and Order of Precedence
- Control Structure- Sequence
- Control Structure – Selection
- Control Structure – Loops

Data Analysis and Problems
- Program Development Process
- Structure Diagrams
- Organising Information
- Fixed and Variable Length Records
- Analysing the Problem
- Determining the Structure of a Program.
- Structured Programming Diagrams
- CRC Cards

Further Programming Techniques
- Program Structures
- Data Structures
- Functions
- Arrays
- Link Lists
- Data Structures: Queues and Stacks
- Data Structures: Graphs and Trees.

Testing
- Software and Testing
- Documentation of Tests. Levels of Testing, Desk Checking and Dry Running
- The Diagnostic Aids Generated During Compilation and Runtime
- The Problem and Techniques of Program Maintenance
- The need for Robust and Reliable Software

Implementation

- Traditional Systems Lifecycle
- The Software Development Lifecycle
- The Need for Documentation and Coding Standards
- Attributes of Good Documentation.
- The Elements of Documentation
- Techniques of Documentation and The Programmer's Role

Prescribed text

NARDI, Bonnie A. (1993) A small matter of programming: Perspectives on end user computing, MIT Press.

Recommended text

SPRANKLE, Maureen (1998) Problem solving and programming concepts, 4th edition, Prentice Hall.

TOMASSO, C. (1971) Computer programming: A practical course in SL/1, Foulsham.